Form AR140

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP5004
Subject Title	Integrated Circuit Processing and Laboratory
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	To make familiar with the microfabrication concepts, materials, and methods that are typically used in a cleanroom especially for the fabrication of integrated circuits (ICs).
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) understand the basic knowledge of different classes of cleanrooms, the working procedures, and the safety aspects; b) understand the principles behind the design and fabrication of IC devices and systems and the effect of processes on their performance; c) have a thorough understanding of the available fabrication technologies; and d) experimentally carry out a simple process recipe using the most common microfabrication techniques for IC processing.
Subject Synopsis/ Indicative Syllabus	• Physical principles of IC fabrication processes; • Surface preparation; • Thermal processes; • Chemical and physical vapor depositions; • Resist coating and removal; • Mask fabrication and advanced lithography; • Etching techniques; • Process characterization;
Teaching/Learning Methodology	In order to stimulate and motivate the students' interest in the study of cleanroom microfabrication technologies for ICs, several cleanroom microfabrication experiments will be offered to the students for them to gain hands-on experience on the growth of SiO_2 thin film by thermal oxidation, CVD, PVD, lithography, patterning and etching for simple ICs. These proposed practical examples will demonstrate the importance of microfabrication in the forefront of modern microelectronics circuits.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
		a	b	c	d
1. Continuous assessments (Assignments and tests)	50	✓	✓	✓	
2. Report & presentation	50	✓	✓	✓	✓
Total	100 %				

Continuous assessments consist of assignments and mid-term tests. These will strengthen the students' basic knowledge and the analytical skill to solve the problems related to cleanroom microfabrication technologies as well as the cleanroom experimental skills on microfabrication.

The report and presentation include a written report and an oral presentation. These will review the students' hands-on skills of some experiments.

Student Study Effort Expected

Class contact:	
▪ Lectures	18 Hrs.
▪ Laboratory	21 Hrs.
Other student study efforts:	
▪ Self-study	81 Hrs.
The total student study effort	120 Hrs.

Reading List and References

- S. Franssila, Introduction to Microfabrication, John Wiley & Sons, 2010.
- J. D. Plummer, M. D. Deal, and P. B. Griffin, Silicon VLSI Technology, 2nd Edition, Prentice Hall, 2017.
- S.Wolf & R.N.Tauber, Silicon Processing for the VLSI Era, vol.1, 2nd edition, Lattice, 2000.
- M. Madou, Fundamentals of Microfabrication, CRC Press, 2002.